

The value of small urban greenspaces for birds in a Mexican city

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ABSTRACT

We explore bird communities in different types of small urban greenspaces (<2 ha) in order to understand which birds occupy these structures, and the habitat and landscape features that affect them. In particular we analyzed how greenspace characteristics (size and vegetation structure), those of the adjacent landscape (area covered by urban structures and vegetation), and human disturbance variables (traffic of pedestrians, vehicles and noise levels) affected bird species richness, abundance and community composition (during summer and winter) in the city of Pachuca, Mexico. We selected 19 small greenspaces (ranging from 0.1 to 2 ha), including public parks, gardens and road strip corridors. We registered 39 species of birds of which 15 were migratory. In general bird species richness was higher in parks and gardens, and lower in road strip corridors, where more noise and traffic was registered. Greenspace area was the most important variable that positively influenced bird species richness, for both the summer and the winter communities. Summer bird species richness was lower in places that had a greater percentage of area covered by buildings in the adjacent landscape. Generalist and opportunistic species were favored by urbanization. Insectivorous species in winter were more abundant in larger greenspaces that had taller trees and more tree and shrub species, and in sites with more vegetative cover in the adjacent landscape. We suggest some management and urban planning actions that would benefit birds within many Latin American cities, where small greenspaces are often the only available refuges for birds.

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1. Introduction

Greenspaces in cities (e.g., woodlots, parks, gardens, road strip corridors, golf courses, and cemeteries) can be important for the conservation of at least some of the original biodiversity in urbanized landscapes (Adams, 1994; Fernández-Juricic and Jokimäki, 2001; Lussenhop, 1977; McDonnell and Pickett, 1990). These areas have been a priority for planners and land managers in many developed nations because they are also important for improving the quality of life in many ways, from the environmental services they provide to the aesthetic, psychological, health and social benefits for human inhabitants (e.g., Azerrad and Nilon, 2001; Barker and Graf, 1989; Chiesura, 2004; Kaplan, 1983). Some studies have demonstrated that greenspaces can sustain a diverse avifauna, although species richness and abundance of birds may depend on many factors such as the size of the greenspace, its vegetation structure, isolation, and human disturbance (reviewed by Fernández-Juricic and Jokimäki, 2001).

Greenspace area has been found to explain a high percentage of variation in total bird species numbers or the occurrence of individual species in many cities (Fernández-Juricic, 2000a, 2004; Jokimäki, 1999; Mörtberg, 2001; Murgui, 2007; Park and Lee, 2000). It has been calculated that urban parks of 10–35 ha would contain most of the species of birds recorded in some cities (Fernández-Juricic and Jokimäki, 2001). However, in the majority of studies larger greenspaces (from 2 to 700 ha) are considered, and small areas (<2 ha) are often ignored, possibly because they are almost pure edge habitat (Collinge, 1996) and some bird species avoid them (Jokimäki, 1999). In many Latin American cities larger greenspaces (>2 ha) are inexistent, and a few small parks and gardens are the only refuges available for birds. Furthermore, the small number of existing greenspaces is disappearing as these sites are considered valuable for the construction of new office buildings, as well as housing and commercial developments (Chacalo et al., 1994; Nilsson et al., 1997).

With an appropriate design and adequate management practices, even small greenspaces could decrease the negative consequences of the chaotic urban development seen in Latin American countries and many other developing nations, where the highest growth of human population is expected in the future. It has been estimated that the urban population in developing nations will reach 3.5 billion by 2025, three times more than

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Fig. 1. Location of the city of Pachuca in the state of Hidalgo. The distribution of the 19 study sites (parks, gardens and road strip corridors) is shown.

the urban population estimated for developed nations (United Nations, 2009). Despite their potential importance, in these countries there is a lack of studies on how greenspaces in cities can benefit local avifauna (Chace and Walsh, 2006; González-Oreja et al., 2007; Marzluff et al., 2001). Particularly for Mexico, there are only a few published studies that refer to birds and urban areas (e.g., González-Oreja et al., 2007; MacGregor-Fors, 2008; Ortega-Álvarez and MacGregor-Fors, 2009), even though biodiversity in Mexico is severely threatened, and alternative management practices are urgently needed especially for urbanized landscapes.

In this study we explore bird communities in different types of small urban greenspaces (i.e., parks, gardens and road strip corridors) in order to understand which birds occupy these structures and the habitat and landscape features that affect them. We conducted our study in the city of Pachuca, Hidalgo, because it represents a typical city of central Mexico, with a very high rate of population growth (up to 9.0% in the period 2000–2005; INEGI, 2008) and with a tendency of becoming, along with Distrito Federal, Puebla, Querétaro, Tlaxcala, Toluca, San Juan del Río, and Cuernavaca, one of the biggest metropolis (with 36.7 million people by 2050: Garza, 2002). In particular we analyzed how greenspace characteristics (size and vegetation structure), those of the adjacent landscape (area covered by urban structures and vegetation), and human disturbance variables (traffic of pedestrians, vehicles and noise levels) affect bird species richness, abundance and community composition. Because bird community composition changes throughout the year in central Mexico, we considered two bird communities according to seasonality and the residence or migratory status of bird species: summer (breeding resident species only, from May to August) and winter (resident, migratory and transients or accidentals, from January to April and from September to December).

2. Methods

2.1. Study area

Pachuca, central Mexico, is located between 19°50' and 20°10'N, and between 98°41' and 98°57'W, at 2400–2800 m above sea level (Fig. 1). Its metropolitan area covers approximately 100 km² and it has a human population of 445,406 inhabitants (INEGI, 2005). Pachuca's metropolitan area is growing very fast, the mean annual population increase rate was 2.2% in 2005 (Bautista-Ramírez et al., 2008), and in a period of 4 years (2000–2004) the urban surface increased 25% (Gómez-Aíza, 2008), while 22,845 new houses were built from 2000 to 2005 (Bautista-Ramírez et al., 2008).

The mean annual temperature is 15 °C and the mean annual precipitation is 367.6 mm; in summer mean temperature reaches 15.5 °C (9.5–25.4 °C) and the wettest months are July and September; in winter the mean temperature is 12 °C (4.2–24.5 °C) and the driest month is December (INEGI, 2005; Pavón and Meza-Sánchez, 2009). The native vegetation consists of small patches of arid tropical scrub ("matorral xerófilo" according to Rzedowski, 1994), represented mainly by several species of cactus (*Opuntia* spp., *Cylindropuntia imbricata*), huisaches (*Acacia* spp.), and native magueyes (*Agave* spp.), and it covers only about 8% of the city area (Gómez-Aíza and Zuria, 2010). Similar to other Latin-American cities, there are four main urban land-use categories in Pachuca: (1) residential, (2) commercial, (3) industrial, and (4) green areas (Fleming, 1999; Ortega-Álvarez and MacGregor-Fors, 2009). The most important land-use type (about 75% of the city area) consists of housing developments (1–2 stories) (Fleming, 1999), and there are no permanent water bodies.

There are around 30 public greenspaces in Pachuca, mostly gardens, and in general they are small (<2 ha), very similar in age, structure, and maintenance practices; for example, most parks

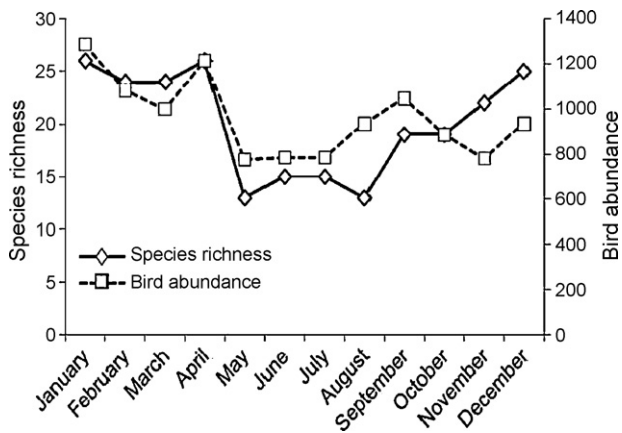


Fig. 2. Overall species richness and total abundance of birds in greenspaces of Pachuca during 2007.

and gardens were established in the second half of the twentieth century, all include many planted non-native trees, a limited shrub stratum, and a continuous maintenance regime applied by the municipal authorities that involves constant lawn mowing, pruning, and removal of dead trees and organic matter (Carbó-Ramírez, 2008). We selected 19 greenspaces (Fig. 2) of three types: parks ($N=6$), gardens ($N=7$) and road strip corridors ($N=6$). These 19 greenspaces are the most important and conspicuous in the city, and can be considered representative of other urban green areas in many Latin American cities, because most parks and gardens in these cities include areas of lawn that are watered and mowed regularly, a shrub cover, a mix of deciduous and coniferous trees, and areas for human recreation, while road strip corridors consist of trees and shrubs planted along streets (Ceballos-Ibarra, 1997; Cobo, 1997; González-Claverán, 1997). In our study, parks included greenspaces designed for human recreation (paved trails for walking or jogging, some include basketball courts and children's playgrounds) with an area ranging between 0.5 and 2 ha, and with a constant pedestrian presence. Gardens included greenspaces designed mainly for decorative purposes with an area generally smaller than 0.5 ha and with very little human presence. Road strip corridors were linear strips of vegetation with variable area (0.2–2 ha) that divided an important street or avenue. All 19 greenspaces can be considered managed areas, and none included patches of native vegetation.

2.2. Habitat variables

For each greenspace we measured habitat characteristics at two spatial scales, considering the site level (e.g., greenspace characteristics: area, perimeter, and vegetation structure), and the adjacent landscape (land use type in 100-m width buffers). At the site level we measured the area and perimeter of each site using a satellite image of the city (2004, IKONOS-2, resolution = 1 m²) and Arcview (ver. 3.2, ESRI). Vegetation structure was evaluated by: (1) tree cover, (2) ground cover, (3) tree and shrub species richness, (4) tree height, (5) shrub height, and (6) herbaceous plant height (for details see Carbó-Ramírez, 2008). Herbaceous species richness was not considered since all greenspaces presented few herbs, mainly grass that was constantly mowed. To estimate habitat variables at the adjacent landscape level, we digitized all greenspaces and other important landscape features, e.g., buildings, roads, and areas covered by native vegetation, from the satellite image. Our habitat identifications were verified by ground truthing (i.e., ground surveys to verify that all landscape features were the same as in the image and that no important changes had occurred since the image was taken in 2004). We calculated distances between the center of

each greenspace and the following structures: closest area covered by native vegetation and closest main road (i.e., more than one lane and constant traffic). A 100-m width buffer was delineated around the border of each greenspace and the following variables were calculated within each buffer: percentage area covered by greenspaces, by asphalt roads, and by buildings. We also used 200-m width buffers, but since results were similar we only included those obtained from 100-m width buffers.

Additionally we obtained three measures related to human disturbance in the 19 greenspaces: number of pedestrians (no. pedestrians/min), number of vehicles circulating in adjacent streets (no. vehicles/min), and noise levels. The number of pedestrians was recorded in a 20-m radius circular plot for three consecutive minutes, at the same locations where bird point counts were conducted (see below), and just after birds were counted. We then calculated the average number of pedestrians at each greenspace, and this number was then divided over the total amount of sampling time (min) for comparative purposes. Noise levels, expressed in A-weighted decibels (dBA), were measured in the adjacent streets on working days from 8 to 10 am under ideal meteorological conditions (no wind and no rain). We used a sound level meter (Extech 407732) and recorded instantaneous noise levels every 10 s for 1 min. Measurements were then averaged to obtain one noise level measure per greenspace. Also, vehicles circulating in adjacent streets were counted for three consecutive minutes. We calculated the average number of vehicles circulating around each greenspace, and this number was then divided over the total amount of sampling time (min) for comparative purposes. These measurements were taken in the early mornings during working days.

2.3. Bird counts

All sites were visited during working days once in the morning (1 h after dawn) and once in the evening (2 h before sunset), every month from January to December 2007. Two types of survey methods were used for counting birds depending on the type of greenspace. For parks and gardens we conducted 5 min unlimited radius point counts (Ralph et al., 1996). Point counts were located at a minimum distance of 100 m from each other in order to assure data independence (Ralph et al., 1996), while greenspaces were separated by at least 200 m. For lineal road strip corridors we used 500 m line transects (Gregory et al., 2004; Ralph et al., 1996). Transects were walked slowly for 10 min recording all birds seen or heard in the transect and within 10 m of the adjacent street at each side of the transect. Depending on the size of the greenspace, we conducted one or two point counts in each area (i.e., one point count for greenspaces smaller than 0.5 ha), and from 1 to 3 line transects at each road strip corridor (i.e., one transect for each 500-m). Data on species richness and abundance were then divided over the total amount of sampling time (min) for comparative purposes. We believe that few birds were undetected regardless of season because of the narrow widths of road strip corridors and the simple vegetation structure of all the greenspaces studied. Each of the two survey methods was selected because it was found appropriate for counting birds in the different types of structures. For example, it was not possible to use line transects in parks and gardens due to their small area. On the other hand, line transects were appropriate for road strip corridors due to their linear nature and narrow width. In order to verify if the two methods were comparable and had the same sampling efficiency we conducted preliminary observations using both survey methods in five road strip corridors, and compared the data using *t*-tests. We found no significant differences in species richness ($t=0.497$, $df=8$, $P=0.633$) or bird abundance ($t=0.395$, $df=8$, $P=0.703$) between the two survey methods, therefore our results are comparable.

Table 1
Greenspace characteristics, adjacent landscape level habitat variables, human disturbance variables, and overall bird species richness and bird abundance for the three types of urban greenspaces in the city of Pachuca, Hidalgo, Mexico. We present averages ($\pm$ standard error).

Habitat variable	Gardens	Parks	Road strip corridors	Statistic $F/H/U^a$	P	Overall interval
Greenspace characteristics						
AREA (ha)	0.3 $\pm$ 0.05 ^a	1.2 $\pm$ 0.30 ^b	1.1 $\pm$ 0.26 ^b	$H = 9.8$	0.008	0.12–2.03
PERIMETER (m)	276.2 $\pm$ 29.43 ^a	522.5 $\pm$ 43.84 ^{ab}	2064.1 $\pm$ 256.66 ^b	$H = 16.0$	<0.001	192.80–3048.10
%CC	45.4 $\pm$ 3.64	54.0 $\pm$ 4.67	64.8 $\pm$ 7.23	$F = 3.5$	0.05	35–87
%GC	53.0 $\pm$ 7.89	55.3 $\pm$ 3.85	69.0 $\pm$ 13.18	$F = 0.9$	0.43	11–100
TSHSPP	3.6 $\pm$ 0.37 ^a	5.5 $\pm$ 0.34 ^b	2.5 $\pm$ 0.56 ^{ac}	$F = 11.9$	<0.001	1–7
TREEH (m)	8.9 $\pm$ 1.70	8.5 $\pm$ 1.86	5.7 $\pm$ 0.99	$F = 1.20$	0.34	2–16
SHRUBH (m)	0.3 $\pm$ 0.18	1.1 $\pm$ 0.21	Not present	$U = 7.0$	0.05	0–2
HERBH (m)	0.2 $\pm$ 0.06	0.1 $\pm$ 0.04	0.0 $\pm$ 0.01	$H = 5.5$	0.06	0.02–0.46
Adjacent landscape						
%CGS	12.2 $\pm$ 2.05 ^a	11.9 $\pm$ 3.08 ^a	28.4 $\pm$ 6.48 ^b	$F = 5.08$	0.02	2.39–53.75
%CASP	39.7 $\pm$ 1.85	47.4 $\pm$ 4.49	42.1 $\pm$ 2.66	$F = 1.6$	0.23	32.59–62.34
%CBUILD	48.1 $\pm$ 3.77	40.1 $\pm$ 6.23	29.5 $\pm$ 7.95	$F = 2.4$	0.12	4.77–61.02
DNATVEG (m)	738.1 $\pm$ 87.28	806.7 $\pm$ 77.66	1209.7 $\pm$ 363.53	$H = 1.5$	0.47	469.00–2955.00
DGREENS (m)	113.0 $\pm$ 17.28	91.3 $\pm$ 31.09	751.9 $\pm$ 418.37	$H = 1.5$	0.48	10.47–2425.34
DMROAD (m)	89.8 $\pm$ 31.52 ^a	5.6 $\pm$ 4.60 ^b	0.0 $\pm$ 0.00 ^c	$H = 16.0$	0.003	0.00–246
Human disturbance						
Pedestrians (min)	12.5 $\pm$ 1.97 ^a	59.2 $\pm$ 12.85 ^b	11.7 $\pm$ 2.85 ^a	$H = 11.3$	0.004	2.83–105.33
Vehicles (min)	8.0 $\pm$ 1.63 ^a	18.8 $\pm$ 2.70 ^b	28.9 $\pm$ 3.32 ^c	$F = 17.3$	<0.001	0.29–42.50
Noise level (dBA)	59.2 $\pm$ 2.36 ^a	63.3 $\pm$ 1.10 ^{ab}	68.0 $\pm$ 0.87 ^b	$H = 8.43$	0.015	49.95–71.41
Bird species richness (#species/min)	0.21 $\pm$ 0.004 ^a	0.13 $\pm$ 0.007 ^a	0.05 $\pm$ 0.004 ^b	$H = 28.81$	<0.001	0.04–0.31
Bird abundance (#birds/min)	3.05 $\pm$ 0.276 ^a	3.34 $\pm$ 0.160 ^a	1.44 $\pm$ 0.128 ^b	$F = 27.06$	<0.001	0.91–5.30

^a Results of statistical tests (ANOVA, Kruskal–Wallis, or Mann–Whitney U) for comparing the three types of greenspaces. Different superscripts show groups statistically different in post hoc tests ($P < 0.05$). AREA = area of the greenspace; PERIMETER = perimeter of the greenspace; %CC = percentage canopy cover; %GC = percentage ground cover; TSHSPP = tree and shrub species richness; TREEH = tree height; SHRUBH = shrub height; HERBH = herbaceous plant height; %CGS = percentage of area covered by greenspaces in 100-m width buffers; %CASP = percentage of area covered by asphalt roads in 100-m width buffers; %CBUILD = percentage of area covered by buildings in 100-m width buffers; DNATVEG = distance to the closest area covered by native vegetation; DGREENS = distance to the closest greenspace; DMROAD = distance to the closest main road.

We classified birds according to their residence status based on Howell and Webb (1995) and Martínez-Morales et al. (2007). Endemisms and status of endangerment were based on the Red List of Threatened Species (IUCN, 2001), SEMARNAT (2002), and González-García and Gómez de Silva (2003). The Taxonomical classification and scientific names are those suggested by the AOU (1998, 2009).

2.4. Data analysis

For the summer and winter communities we obtained species richness at each greenspace surveyed and total abundance of birds. We classified birds into trophic groups according to Howell and Webb (1995), MacGregor-Fors (2008), and Ortega-Álvarez and MacGregor-Fors (2009).

To assure that our survey effort was enough to record a representative sample of bird communities, we constructed species accumulation curves (observed and estimated) for each site and overall. Species accumulation curves and non-parametric richness estimators (Chao 1 and Chao 2) were calculated using EstimateS 7.5 (Colwell, 2004).

We checked for normality and homogeneity of variances in the data by means of a Kolmogorov–Smirnov test and a Levene Median test, respectively. We compared greenspace characteristics, adjacent landscape level habitat variables, and human disturbance variables between parks, gardens and road strip corridors using ANOVA and Tukey post hoc tests. When assumptions of normality and homogeneity of variances were violated, we used Kruskal–Wallis or Mann–Whitney U tests, and Bonferroni post hoc tests. Linear multiple regression (Kleinbaum et al., 1998) was used to analyze the relationship between species richness and bird abundance and the explanatory variables for both communities. Our main objective was to identify the subset of explanatory variables which accounted for a biologically important proportion of the variation in the bird population parameters, therefore our final models were the result of a non-automatic variable search based on forward elimination techniques. Explanatory variables were found to

be statistically significant for explaining the variation in a specific Y variable when $P < 0.05$. The residuals of the final models obtained with linear multiple regression revealed no obvious suggestions of non-normality or collinearity (Kleinbaum et al., 1998). Statistical analyses were performed using SigmaStat (Jandel Scientific, 2006, Version 3.5) and SAS (Version 8.00, SAS Institute Inc., 1999). To avoid the mistake of mismatching different spatial scales within a single analysis (Morrison et al., 1998; Wiens, 1989), greenspace characteristics and habitat variables measured in the adjacent landscape were analyzed separately.

Canonical correspondence analysis (CCA) was used to examine relationships between both bird communities (summer and winter) and habitat variables (TerBraak, 1986). Each CCA included between 17 (summer community) and 37 species (winter community), and 19 greenspaces. Species data included the total number of individuals (all surveys together) for each species observed at each greenspace. Global permutation tests (carried out by Monte Carlo permutation tests with 199 permutations under the reduced model) were performed to judge the statistical significance of the relationship between the species and habitat variables. CCA analyses were performed using Canoco for Windows (Version 4.0, TerBraak, 1998). Species data were first logarithmically transformed (base 10) after adding a constant because of the presence of zeroes in these data; all data were included in the analyses and we used the option for downweighting rare species (TerBraak, 1986).

3. Results

3.1. Habitat characteristics

Urban greenspaces in the city of Pachuca measured from 0.12 ha to 2.03 ha (Table 1). Gardens were significantly smaller than parks and road strip corridors, and had a smaller perimeter than road strip corridors (Table 1). Road strip corridors had fewer tree and shrub species than parks, and did not present a shrub stratum (Table 1). At the adjacent landscape level, we only found differences in the percentage cover of greenspaces in 100-m width buffers, and in

Table 2

Tree and shrub species present in gardens, parks and road strip corridors in the city of Pachuca, Hidalgo, Mexico. Numbers indicate the number of greenspaces where the species was recorded.

Family	Species	Greenspaces		
		Gardens	Parks	Road strip corridors
Agavaceae	<i>Yucca filifera</i>	1	2	1
Anacardiaceae	<i>Schinus molle</i>	0	1	3
Bignoniaceae	<i>Jacaranda</i> sp.	0	1	0
Buxaceae	<i>Buxus</i> sp.	2	3	0
Cupressaceae	<i>Cupressus</i> spp.	0	1	0
	<i>Juniperus</i> spp.	4	3	1
	<i>Thuja</i> sp.	0	1	0
Casuarinaceae	<i>Casuarina equisetifolia</i>	1	2	2
Leguminosae	<i>Erythrina coralloides</i>	1	0	0
Myrtaceae	<i>Eucalyptus camaldulensis</i>	2	1	1
	<i>Eugenia paniculata</i>	0	1	0
Nyctaginaceae	<i>Bougainvillea</i> sp.	0	1	0
Oleaceae	<i>Fraxinus uhdei</i>	4	4	2
	<i>Ligustrum japonicum</i>	0	1	0
	<i>Ligustrum lucidum</i>	2	4	0
Palmaceae	<i>Phoenix</i> sp.	1	0	2
Pinaceae	<i>Pinus</i> spp.	4	4	2
Salicaceae	<i>Populus</i> sp.	2	2	1
Taxodiaceae	<i>Taxodium macronatum</i>	1	1	0
Total species		12	17	9

the distance from the green area to the closest main road (Table 1). In general, the landscape matrix around road strip corridors contained a greater percentage of area covered by greenspaces than the landscape matrix around parks and gardens. Gardens were located at the greatest distance from a main road, followed by parks, while road strip corridors were next to a main road. Overall, gardens were less disturbed than parks and road strip corridors. We observed significantly more visitors (pedestrians/min) in parks, and more noise and traffic in road strip corridors and parks (Table 1).

In general, *Pinus* spp., *Fraxinus uhdei*, and *Juniperus* spp. were detected most often in the urban greenspaces of Pachuca (Table 2). Although we found only from 1 to 7 tree and shrub species per greenspace (Table 1), overall parks presented more tree and shrub species, followed by gardens and road strip corridors (Tables 1 and 2).

3.2. Bird surveys

We registered 39 species from 18 families and 33 genera (Table 3). Of them, 22 species were resident, 15 were considered migratory birds, one was transient and we could not determine the residence status of one species that could not be identified to the species level (Table 3). We also registered barn swallows (*Hirundo rustica*) flying high above the sites but we did not include them in our analyses, since there was no evidence that indicated these birds were actually using the greenspaces for breeding or feeding. None of the recorded species are considered endangered by the IUCN (2001) and only one (*Oporornis tolmiei*) is considered as threatened by the Mexican government (SEMARNAT, 2002). One species (*Turdus rufopalliatus*) is endemic to northwest Mexico, and its presence in our study site constitutes the first record for the state of Hidalgo (Carbó-Ramírez, 2008; Martínez-Morales et al., 2010); also three species are semiendemic and two are quasi-endemic to Mexico (according to Howell and Webb, 1995). Bird species recorded represent five trophic groups: 59% of the species are insectivores, 28% are granivores, 5% are nectarivores, 5% are frugivores, and 3% are omnivores (Table 3).

For the summer community, we registered 17 species. Five species (*Bubulcus ibis*, *Lampornis clemenciae*, *Picoides scalaris*, *Mimus polyglottos* and *Melospiza melodia*) that were considered resident

according to Howell and Webb (1995) and Martínez-Morales et al. (2007) (see Table 3) were not observed during our summer surveys. Non-parametric richness estimators (Chao 1 and Chao 2) calculated for this community indicated that the inventory was complete. For 84% of the sites, completeness was over 80%, so we consider that our survey effort was appropriate. For the winter community we registered 37 species, and the analysis indicated that the inventory was not complete with 3–4 species missing. For this community, only in 53% of the sites completeness was over 80%, mainly due to the large number of migratory, transient or accidental species recorded during winter.

Total species richness and bird abundance in greenspaces were variable over the months. Species richness was lower from May to August ($N=14$) and higher from January to April ($N=25$) (Fig. 2). Bird abundance was the lowest in May ($N=778$) and highest in January ($N=1285$) (Fig. 2). In general, species richness and abundance were higher in parks and gardens, and lower in road strip corridors (Table 1).

3.3. Influence of habitat variables on avifauna

3.3.1. Summer community

At the site level (greenspace characteristics), we found a significant and positive relationship between species richness and greenspace area (Table 4), and between species richness and the height of herbaceous plants (Table 4). No other relationship was found significant. At the adjacent landscape level, we found a negative relationship between species richness and the percentage of area covered by buildings in 100-m width buffers around the greenspaces (Table 4). We did not obtain any significant models with the human disturbance variables.

The global permutation tests based on the sum of all canonical eigenvalues indicated that the relationship between species and greenspace characteristics, and between species and human disturbance variables, were not significant ($P>0.05$). However, it was significant for the adjacent landscape-level variables measured in 100-m width buffers ($P=0.002$). The first two axes explained 29.2% of the variance in bird species, and 100% of the variation in the species–habitat relationship (Table 5). In general, species like *Columba livia* (C.liv), *Passer domesticus* (P.dom) and *Molothrus ater* (M.ate) were favored by urbanization; while species like *Zenaidura macroura* (Z.mac) and *Carpodacus mexicanus* (C.mex) were more abundant in areas of the city with more vegetative cover (Fig. 3).

3.3.2. Winter community

At the site level we only found a positive and significant relationship between winter species richness and the area of the sites (Table 4), and we did not obtain any significant models that explained total bird abundance. The landscape analysis revealed a positive relationship between species richness and the percentage of area covered by asphalt in 100-m width buffers (Table 4), and we could not obtain any significant models that explained bird abundance. Also, none significant models were obtained with the human disturbance variables.

In winter, we found a significant relationship between species and greenspace characteristics ($P=0.005$). The first two axes explained 22.4% of the variance in bird species, and 60.7% of the variation in the species–habitat relationship (Table 5). In general, insectivorous species like *Sphyrapicus varius* (S.var), *P. scalaris* (P.sca), *Tyrannus vociferans* (T.voc), *Poliophtila caerulescens* (P.cae), *Mniotilta varia* (M.var), and *Piranga ludoviciana* (P.lud), as well as frugivorous like *T. rufopalliatus* (T.ruf), were more abundant in larger greenspaces that had taller trees and more tree and shrub species (Fig. 4). Some species were more abundant in open areas like *Toxostoma curvirostre* (T.cur), *Sturnus vulgaris* (S.vul), *Quiscalus*

Table 3
Total abundances of birds recorded in three types of urban greenspaces in the city of Pachuca, Hidalgo, Mexico. Residence status: R = breeding resident; M = winter migrant; T = transient. Species codes are used in ordination diagrams.

Family	Species codes	Species ^a	Greenspaces			Residence status
			Gardens	Parks	Road strip corridors	
Ardeidae	B.ibi	<i>Bubulcus ibis</i> (I)	0	0	36	R
Columbidae	C.liv	<i>Columba livia</i> (G)	584	1632	531	R
	Z.mac	<i>Zenaida macroura</i> (G)	0	0	3	R
	C.inc	<i>Columbina inca</i> (G)	94	232	98	R
Trochilidae	C.lat	<i>Cyanthus latirostris</i> (N) ^c	21	17	6	R
	L.cle	<i>Lampornis clemenciae</i> (N) ^c	1	0	0	R
Picidae	S.var	<i>Sphyrapicus varius</i> (I)	0	4	0	M
	P.sca	<i>Picoides scalaris</i> (I)	0	1	0	R
Tyrannidae	–	<i>Contopus</i> sp. (I)	0	1	0	–
	P.rub	<i>Pyrocephalus rubinus</i> (I)	23	9	14	R
Troglodytidae	T.voc	<i>Tyrannus vociferans</i> (I) ^d	12	5	6	R
	T.bew	<i>Thryomanes bewickii</i> (I)	17	31	3	R
Regulidae	R.cal	<i>Regulus calendula</i> (I)	24	17	6	M
Sylviidae	P.cae	<i>Polioptila caerulea</i> (I)	3	2	1	M
Turdidae	T.ruf	<i>Turdus rufopalliatu</i> (F) ^b	19	139	3	R
Mimidae	M.pol	<i>Mimus polyglottos</i> (I)	1	0	0	R
	T.cur	<i>Toxostoma curvirostre</i> (I)	19	28	30	R
Sturnidae	S.vul	<i>Sturnus vulgaris</i> (G)	0	2	18	R
Bombacillidae	B.ced	<i>Bombacilla cedrorum</i> (F)	208	284	421	M
Parulidae	V.cel	<i>Vermivora celata</i> (I)	3	0	0	M
	V.ruf	<i>Vermivora ruficapilla</i> (I)	2	13	0	M
	D.cor	<i>Dendroica coronata</i> (I)	140	260	76	M
	D.tow	<i>Dendroica townsendi</i> (I)	0	1	1	M
	M.var	<i>Mniotilta varia</i> (I)	0	1	0	M
	O.tol	<i>Oporornis tolmiei</i> (I) ^e	0	2	0	M
	W.pus	<i>Wilsonia pusilla</i> (I)	21	18	9	M
	P.rubr	<i>Piranga rubra</i> (I)	1	0	0	M
	P.lud	<i>Piranga ludoviciana</i> (I)	5	10	0	M
	P.fus	<i>Pipilo fuscus</i> (G)	9	42	15	R
Icteridae	M.mel	<i>Melospiza melodia</i> (G)	0	1	2	R
	Q.mex	<i>Quiscalus mexicanus</i> (O)	145	534	1115	R
	M.aen	<i>Molothrus aeneus</i> (G)	23	5	89	R
	M.ate	<i>Molothrus ater</i> (G)	0	78	17	R
	I.gal	<i>Icterus galbula</i> (I)	2	1	1	M
	I.bul	<i>Icterus bullockii</i> (I) ^c	1	0	0	M
	I.cuc	<i>Icterus cucullatus</i> (I) ^d	1	1	0	T
	C.mex	<i>Carpodacus mexicanus</i> (G)	168	455	610	R
Fringillidae	S.psa	<i>Spinus psaltria</i> (G)	22	61	45	R
	P.dom	<i>Passer domesticus</i> (G)	1006	929	985	R

^a Trophic group: (I) insectivore; (G) granivore; (F) frugivore; (N) nectarivore; (O) omnivore.

^b Endemism: endemic to northwest Mexico.

^c Endemism: semiendemic to Mexico.

^d Endemism: quasi-endemic to Mexico.

^e Conservation status: threatened according to SEMARNAT (2002).

Table 4
Final models and R^2 values obtained from linear multiple regression analyses between bird species richness or bird abundance and the greenspace characteristics and adjacent landscape-level explanatory variables (X). All regression coefficients given are significant at $P < 0.05$. AREA = area of the greenspace; HERBH = herbaceous plant height; %CASP100 = percentage of area covered by asphalt roads in 100-m width buffers; %CBUILD = percentage of area covered by buildings in 100-m width buffers.

Dependent variable	Model	R^2	P
Greenspace characteristics			
Species richness summer community	$5.29 + 1.99\text{AREA} + 6.42\text{HERBH}$	0.65	<0.001
Species richness winter community	$12.98 + 2.06\text{AREA}$	0.20	0.05
Adjacent landscape level			
Species richness summer community	$9.96 - 0.06\%\text{CBUILD}$	0.25	0.03
Species richness winter community	$3.43 + 0.26\%\text{CASP}$	0.45	0.002

mexicanus (Q.mex), *Molothrus aeneus* (M.aen), *M. ater* (M.ate), and *C. mexicanus* (C.mex).

We also found a significant relationship between species and adjacent landscape-level habitat variables for 100-m width buffers ($P = 0.026$). The first two axes explained 22.4% of the variance in bird species, and 100% of the variation in the species–habitat relationship (Table 5). Generalist and opportunistic species like *C. livia* (C.liv), *Columbina inca* (C.inc), *Pipilo fuscus* (P.fus), *Q. mexicanus* (Q.mex), *M. ater* (M.ate), *C. mexicanus* (C.mex), and *P. domesticus* (P.dom) were favored by urbanization (Fig. 5). Some insectivorous and Neotropical migratory birds were more abundant in sites

with more vegetative cover in the adjacent landscape like *S. varius* (S.var), *Pyrocephalus rubinus* (P.rub), *T. vociferans* (T.voc), *Bombacilla cedrorum* (B.ced), *Vermivora celata* (V.cel), *Piranga rubra* (P.rubr), and *Icterus galbula* (I.gal) (Fig. 5). None significant relationships were found between species and human disturbance variables.

4. Discussions and conclusions

In general, greenspace area was the most important variable that positively influenced bird species richness in the city of Pachuca, for

Table 5

Results of CCA analyses for greenspace characteristics and adjacent landscape-level habitat variables. The percentage variance in the species data and species–environment relationship explained by the canonical axes are cumulative.

	Canonical axes			
	1	2	3	4
Summer community				
Adjacent landscape level				
Eigenvalues	0.103	0.048	0.032	0.167
Species–environment correlations	0.873	0.696	0.661	0.000
% variance species data	13.0	19.0	23.0	44.0
% variance species–environment relationship	56.6	82.7	100.0	0.0
Winter community				
Greenspace characteristics				
Eigenvalues	0.111	0.051	0.043	0.037
Species–environment correlations	0.917	0.894	0.818	0.801
% variance species data	15.4	22.4	28.3	33.4
% variance species–environment relationship	41.6	60.7	76.8	90.6
Adjacent landscape level				
Eigenvalues	0.100	0.058	0.045	0.030
Species–environment correlations	0.895	0.801	0.806	0.677
% variance species data	13.9	21.9	28.1	32.3
% variance species–environment relationship	43.1	67.8	87.1	100.0

both the summer and the winter communities. Similar results have been found before in cities of Europe (Cornelis and Hermy, 2004; Fernández-Juricic, 2000a, 2004; Fernández-Juricic and Jokimäki, 2001; Jokimäki, 1999; Mörtberg, 2001; Mörtberg and Wallentius, 2000; Murgui, 2007), North America (Gavareski, 1976), and Asia (Park and Lee, 2000), although in most studies smaller greenspaces (<2 ha) were not considered. Even if only small areas are available, greenspace size must be maximized within cities to favor rich and abundant communities of resident and migratory birds. Also important was the height of the herbaceous stratum during summer, which favored many resident and granivore species, probably by providing them with seeds and some invertebrates. Parks and gardens should include an area with a more diverse herbaceous cover that is not constantly mowed.

Some habitat variables measured in the adjacent landscape affected species richness and abundance of birds found in the greenspaces of Pachuca. In our study, summer bird species richness was lower in places that had a greater percentage of area covered by

buildings in the adjacent landscape. It has been reported before that urbanization diminishes bird species richness (Chace and Walsh, 2006; Marzluff et al., 2001; Ortega-Álvarez and MacGregor-Fors, 2009), however some studies have shown that species richness might peak at lower or intermediate levels of urbanization, rather than at zero or very low urbanization (Gómez-Aíza and Zuria, 2010; Jokimäki and Suhonen, 1993). Many species, especially insectivores, appear to be selecting greenspaces within less urbanized areas of the city. Also, some migratory birds (e.g., *S. varius*, *B. cedrorum*, *V. celata*, *P. rubra*, *I. galbula*, and *I. bullockii*) were mainly found in greenspaces that were located in areas of the city with a greater percentage of vegetative cover (Fig. 5). During winter, bird abundance increased in greenspaces that had a greater percentage of area covered by asphalt roads in the adjacent landscape, but it was mainly due to the common granivores like doves (*C. livia* and *C. inca*), sparrows (*P. domesticus*), and cowbirds (*M. ater*). Roads

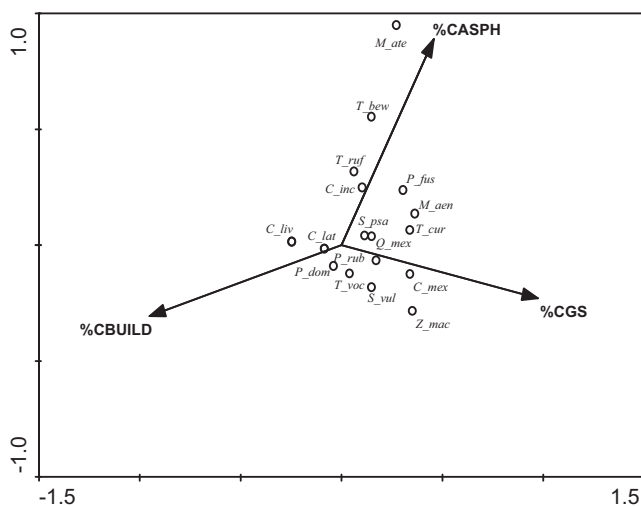


Fig. 3. Environmental gradients of adjacent landscape-level variables and locations of resident bird species in the ordination space constructed with CCA for the summer community (2007). The length and direction of the arrows indicate the relative degree and direction of association. Species codes are in Table 3. %CGS = percentage of area covered by greenspaces in 100-m width buffers; %CASP = percentage of area covered by asphalt roads in 100-m width buffers; %CBUILD = percentage of area covered by buildings in 100-m width buffers.

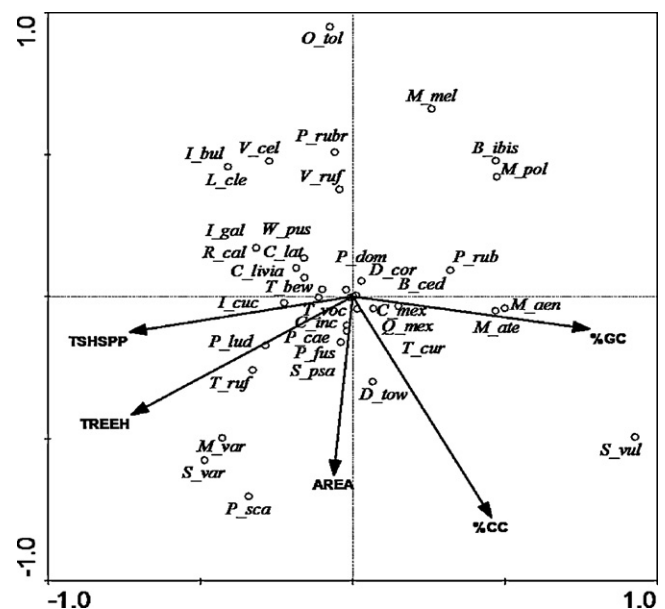


Fig. 4. Environmental gradients of greenspace-level variables and locations of resident and Neotropical migratory birds in the ordination space constructed with CCA for the winter community (2007). The length and direction of the arrows indicate the relative degree and direction of association. Species codes are in Table 3. AREA = area of the greenspace; %CC = percentage canopy cover; %GC = percentage ground cover; TSHSP = tree and shrub species richness; TREEH = tree height.

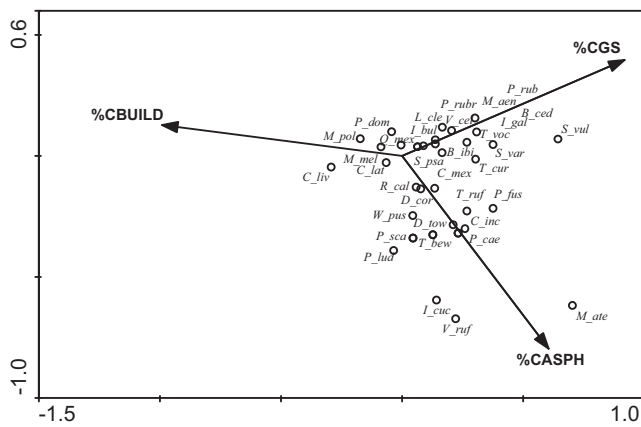


Fig. 5. Environmental gradients of adjacent landscape-level variables and locations of resident and Neotropical migratory birds in the ordination space constructed with CCA for the winter community (2007). The length and direction of the arrows indicate the relative degree and direction of association. Species codes are in Table 3. %CGS = percentage of area covered by greenspaces in 100-m width buffers; %CASP = percentage of area covered by asphalt roads in 100-m width buffers; %CBUILD = percentage of area covered by buildings in 100-m width buffers.

may alter avian community composition by attracting edge species (Miller et al., 2001). For more specialized birds, roads may become sinks by increasing mortality rates (Mumme et al., 2000). However, if carefully designed and managed, streets that connect urban greenspaces, along with vegetation structure, can positively influence bird species richness, species temporal persistence, density of feeding and nesting guilds, and probability of street occupation by individual species (Fernández-Juricic, 2000b).

Human disturbance is another factor that influences bird species richness, abundance and community composition in greenspaces (Fernández-Juricic, 2000c; Fernández-Juricic et al., 2001). In general, we observed lower values of the human disturbance variables in gardens, and these areas had on average more bird species (Table 1). Although parks were heavily visited by people (e.g., up to 105 pedestrians/min, Table 1) they also supported a diverse bird community. In our study place people do not regularly feed birds, but some species (e.g., doves and sparrows) take advantage of human waste. Parks had a more complex habitat structure, and many birds may show more tolerance to people since they have more available cover to hide from visitors (Fernández-Juricic et al., 2001; Knight and Temple, 1995). Noise levels and traffic of vehicles were usually higher on road strip corridors where species richness and bird abundance were the lowest (Table 1). It has been demonstrated that, for some species, habitat quality decreases near roads due to disturbance by car traffic and elevated noise levels (Fernández-Juricic, 2000b; Reijnen and Foppen, 1994; Reijnen et al., 1995).

There are other habitat variables that can influence bird species richness and abundance in greenspaces that were not measured in this study. For example, age of the greenspace has been shown to influence bird species richness (Fernández-Juricic, 2000a). However, the parks and gardens that we considered have approximately the same age because most of them were established in the second half of the twentieth century. Road strip corridors probably have different ages since road construction and modernization is one of the main priorities of the current administration. Connectivity between greenspaces must also influence bird species richness and abundance of birds in cities (Fernández-Juricic, 2000b; Fernández-Juricic and Jokimäki, 2001; Jokimäki, 1999). We did not directly measured connectivity but according to our observations greenspaces in many Mexican cities are mainly isolated. Some bird species may be able to move between greenspaces by flying above buildings and roads, while other species may be more reluctant to

cross open and disturbed spaces (Clevenger et al., 2003; Desrochers and Hannon, 1997), and may be using wooded streets as corridors or as an alternative habitat for feeding and nesting (Fernández-Juricic, 2000b, 2001). In Mexican cities, linear road strip corridors could function as movement corridors between parks and gardens if these areas are well designed, however this is not a priority for local authorities and these sites are often removed or maintained at the minimum possible width to allow more space for traffic. Besides wooded streets, small greenspaces could play a significant role in the connectivity of urban landscapes because they could function as stepping stones which could be temporarily used by different species while moving through urban landscapes (Fernández-Juricic and Jokimäki, 2001).

We found different patterns between summer and winter bird communities, and most studies never consider this temporal variation. During summer, total species richness and bird abundance was lower (Fig. 2) because only resident birds are present in the city. However, many resident species breed in urban greenspaces (Carbó-Ramírez, 2008), and adequate conditions must be present for successful nesting. In winter, species richness and abundance of birds increased because of the presence of Neotropical migrants. Even though some of these birds may be just passing by, others may spend the entire winter in urban greenspaces. Transient birds may use these sites for resting and feeding (Duarte, 2001). More studies are needed on the nesting biology of resident birds (Zuria and Rendón-Hernández, 2010), and on the habitat requirements of Neotropical migrants wintering in Latin America, especially studies that focus on birds using urban greenspaces. We suggest that both communities should be analyzed in further studies and taken into account in management plans. Also, it is necessary to determine how successful birds are in urban greenspaces, because parks and gardens may be attractive to some birds, but they may be functioning as population sinks due to increased mortality (Fernández-Juricic and Jokimäki, 2001; Fernández-Juricic and Tellería, 1999; Jokimäki and Huhta, 2000).

Even small greenspaces in cities can contribute to increase the regional diversity (gamma or total bird diversity in the city), because sometimes these areas include bird species that are not present in the remnant patches of native vegetation. For example, eight of the species that we observed in urban parks and gardens (*T. rufopalliat*, *Vermivora ruficapilla*, *Dendroica townsendi*, *Oporornis tolmiei*, *P. rubra*, *P. ludoviciana*, *Melospiza melodia*, and *Icterus cucullatus*) have not been reported before as part of the avifauna present in the native vegetation (arid tropical scrub), in and around the city (Zuria et al., 2009). Many of these species are associated with a well-developed tree stratum present only in parks and gardens within the city.

Urban planners need better information about the factors affecting wildlife communities in order to create and/or maintain biodiversity in urban ecosystems (Clergeau et al., 2001). Since urban areas will continue to expand in the future, especially in developing nations, we urgently need management plans that consider the temporal and spatial variation of animal communities within the cities, and the environmental factors that influence them. If bird populations can survive in urban areas, the chances of species extinction at the regional level will decrease (Fernández-Juricic, 2004), the aesthetic values of urban greenspaces will be enhanced and public awareness about animal life will be promoted (Savard et al., 2000). We suggest three general management and urban planning actions for developing nations: (1) urban greenspaces must be considered a priority in development plans for the benefit of citizens and wildlife conservation; (2) greenspaces should have the largest possible area and contain a complex vegetative cover, that would support a diverse bird community; and (3) the landscape matrix around greenspaces should be considered and the amount of vegetative cover and connectivity between sites should

be maximized, taking advantage of well designed road strip corridors that will allow movement between sites. These actions will certainly benefit birds and other wildlife within cities, and will create a healthier environment for human inhabitants as well.

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